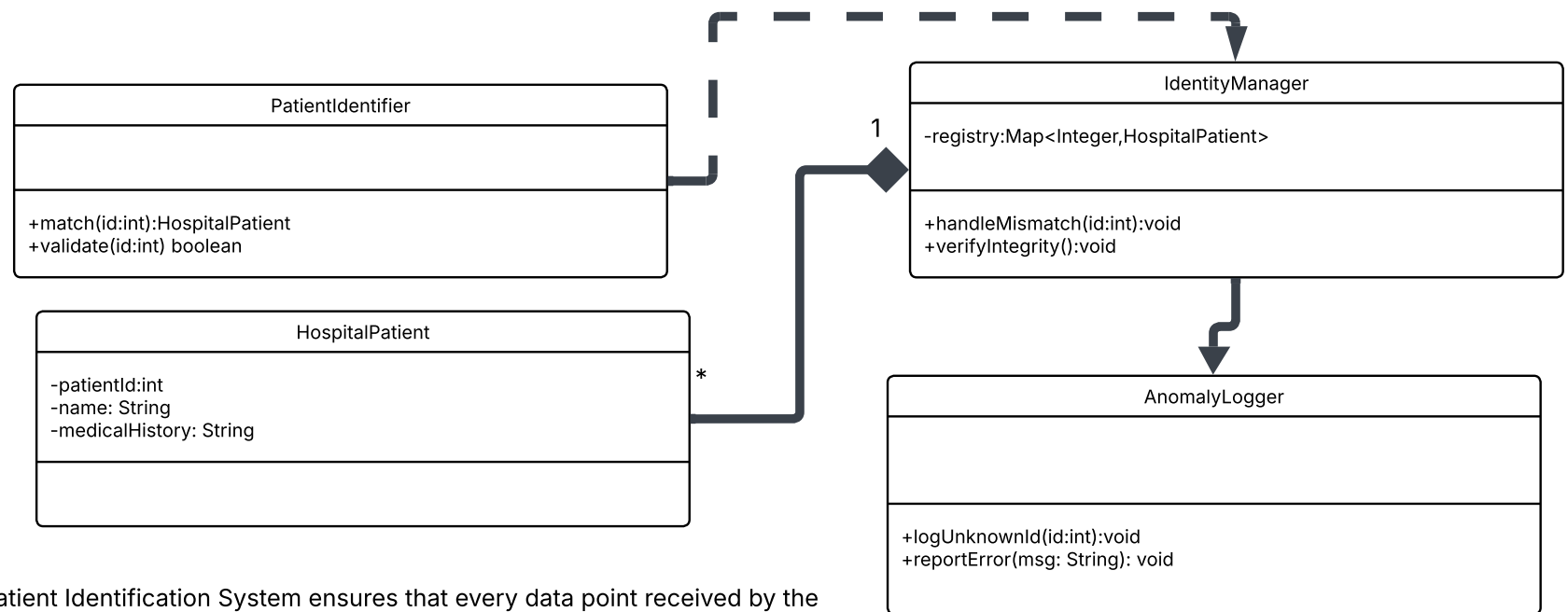




The Patient Identification System ensures that every data point received by the simulator is accurately linked to a specific, verified hospital record. The design focuses on maintaining a reliable connection between raw data streams and actual patient profiles while providing a structured way to handle errors. The PatientIdentifier serves as the primary interface for this process. It receives incoming IDs and uses the match and validate methods to verify if a patient exists in the system. To perform these checks, it communicates with the IdentityManager, which maintains the master registry of all HospitalPatient objects. The use of a composition relationship here indicates that the manager is responsible for the existence and integrity of these patient profiles. A core requirement of this subsystem is the ability to handle mismatches or missing information. When the system receives an ID that cannot be found in the registry, the IdentityManager executes its handleMismatch logic. Instead of allowing the data to be processed incorrectly, the system directs the information to the AnomalyLogger. This class records the unknown ID and generates an error report, ensuring that every failed identification attempt is documented for later review. This structure ensures that only data with a verified link to a registered patient is allowed to enter the monitoring and storage systems.